

- Q.17 Write short note on two-pressure valve.
- Q.18 Define Pascals law, explain the working of any one device working on that principle.

SECTION-D

Note: Long answer type questions. Attempt any one question out of two questions. (1x10=10)

- Q.19 List out various control valve accessories. Explain any three-control valve accessories with figure and its role in control valve.
- Q.20 Make a circuit diagram for a press logic having fixture for a job. Use a suitable cylinder or motor, valve/NRV, and energy supply element. The job is held in fixture, press tool is operated and then the job is ejected.

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**3rd Year /Advance Diploma in
Tool and Die Making
Subject:- Hydraulics and Pneumatics**

Time : 3Hrs.

M.M. : 50

SECTION-A

Note: Multiple choice questions. All questions are compulsory (5x1=5)

- Q.1 Which system posses the lowest weight to power ratio?
- a) Pneumatic system b) Hydraulic system
c) Mechanical system d) Electrical system
- Q.2 Rotary motion in a hydraulic power unit is achieved by using _____.
- a) Hydraulic cylinder b) Pneumatic cylinder
c) Both hydraulic and pneumatic cylinder
d) None of the given

Q.3 In which type of system, power transmission takes place through compressed air?

- a) Fluid power system b) Hydraulic system
- c) Pneumatic system d) Stepper motors

Q.4 Which of the following is a basic component of hydraulic system?

- a) Pump b) Filter
- c) Actuator d) All of the above

Q.5 $\text{Ns/m}^2 = \underline{\hspace{2cm}}$ Poise.

- a) 1 b) 10
- c) 100 d) 1000

SECTION-B

Note: Objective type questions. All questions are compulsory. (5x1=5)

Q.6 Automation

Q.7 Pascal's law

Q.8 Air regulator

Q.9 Combination Valve

Q.10 Linear motion actuator

SECTION-C

Note: Short answer type questions. Attempt any six questions out of eight questions. (6x5=30)

Q.11 Write the advantages of hydraulic system over Pneumatic system.

Q.12 A liquid has a specific gravity of 1.8 and kinematic viscosity of 5.8 strokes. What is its dynamic viscosity?

Q.13 Explain the importance of filtering, pressure regulation and lubrication unit in a pneumatic system.

Q.14 Explain the criteria for the selection of control valves.

Q.15 Draw the neat diagram and explain pneumatic hydraulic actuator in detail.

Q.16 Briefly describe the necessity of FRL unit in pneumatic circuits.